The signal equation of MRI is given by

$$S_c(t) = \int_V s_c(\mathbf{r})\rho(\mathbf{r},t)e^{-i\gamma\int_0^t \Delta B(\mathbf{r},\tau)d\tau}d\mathbf{r} \tag{1}$$

The difference ΔB from the main magnetic field B_0 can be expressed like

$$\Delta B(\mathbf{r},t) = \Delta B_0(\mathbf{r}) + \sum_{\beta\in\{x,y,z\}} G_\beta^0(t)r_\beta + \sum_{\substack{|l|,|m|\geq 0\\ \beta\in\{x,y,z\}}} G_\beta^{lm}(t)N_l^m(\mathbf{r}) \tag{2}$$

Where $N_l^m(\mathbf{r})$ are scalar fields describing the non-linear spatial variations of the magnetic field, typically described either by the solid harmonics

$$R_l^m(\mathbf{r}) = \|\mathbf{r}\|^l Y_l^m(\mathbf{r}), \text{ for spherical harmonics } Y_l^m(\mathbf{r}) \tag{3}$$

or by the

In the ideal case, the higher order terms from the harmonics are zero, and only the linear terms $G_\beta^0(t)r_\beta$ remain. However, in practice there are always some higher order terms present, and the ideal gradient fields precsribed are the actual gradient fields realized. The actual gradient waveforms can be modelled well by a convolution of the ideal gradient waveforms with an impulse response function $\alpha(t)$. This can be expressed for all orders of the gradient fields as

$$G_\beta^{lm}(t) = (\alpha_\beta^{lm} * G_\beta^{\text{ideal}})(t) \tag{4}$$

Or if gradient heating is to be taken into account, the impulse response function can be made time-varying as

$$G_\beta^{lm}(t) = (\alpha_\beta^{lm}(t) * G_\beta^{\text{ideal}})(t). \tag{5}$$

While the magnetization $\rho(\mathbf{r},t)$ is time dependent, we consider readout to be short enough that the only time dependence is due to T_2 decay. Thus, we can express the magnetization as

$$S_c(t) = \int_V s_C(\mathbf{r})\rho(\mathbf{r})e^{-t/T_2(\mathbf{r})}e^{-i\gamma\int_0^t \Delta B(\mathbf{r},\tau)d\tau}d\mathbf{r} \tag{6}$$

With this we can gather all terms that makes the signal equation deviate from a Fourier transform of the static magnetization in the linear gradient fields and form

$$S_c(t) = \int_V s_C(\mathbf{r})\rho(\mathbf{r})e^{\Phi(\mathbf{r},t)}e^{-i\mathbf{k}(t)\cdot\mathbf{r}}d\mathbf{r} \tag{7}$$

where

$$\Phi(\mathbf{r},t) = -t/T_2(\mathbf{r}) - i\gamma\int_0^t \Delta B_0(\mathbf{r})d\tau + \sum_{\substack{|l|,|m|>0\\ \beta\in\{x,y,z\}}} N_l^m(\mathbf{r})\int_0^t G_\beta^{lm}(\tau)d\tau \tag{8}$$

$$\mathbf{k}(t) = \gamma\int_0^t (G_x^0(\tau), G_y^0(\tau), G_z^0(\tau))d\tau \tag{9}$$

Considering a voxel discretization, we can express the operators as matrices by

$$\begin{aligned}
(\mathbf{A}_c)_{kj} &= s_C(\mathbf{r}_j)e^{\Phi(\mathbf{r}_j,t_k)}e^{-i\mathbf{k}(t_k)\cdot\mathbf{r}_j}\\
(\mathbf{A}_c^H)_{ik} &= s_C^*(\mathbf{r}_i)e^{\Phi^*(\mathbf{r}_i,t_k)}e^{i\mathbf{k}(t_k)\cdot\mathbf{r}_i}\\
(\mathbf{A}_c^H\mathbf{A}_c)_{ij} &= \sum_k s_C(\mathbf{r}_j)s_C^*(\mathbf{r}_i)e^{\Phi(\mathbf{r}_j,t_k)+\Phi^*(\mathbf{r}_i,t_k)}e^{-i\mathbf{k}(t_k)\cdot(\mathbf{r}_j-\mathbf{r}_i)}
\end{aligned}$$

1 Approximations for fast application of the normal operator

To get back a Fourier transform, and to enable the Toeplitz property in $\mathbf{A}_c^H \mathbf{A}_c$. We introduce approximations

$$\exp(\Phi(\mathbf{r}_j, t_k)) \approx \sum_{l=0}^{L} \Omega_{kl} \Upsilon_{jl} \tag{10}$$

This enables us to approximate the forward and backward operators by evaluating L fourier transforms. We wish to get a similar decomposition for the normal operator. However, a naive application of the above approximation would give us a decomposition with L^2 Fourier transforms, which is infeasible. We thus try to get a decomposition into fewer Fourier terms. We begin with the naive multiplication

$$\exp(\Phi(\mathbf{r}_j, t_k) + \Phi^*(\mathbf{r}_i, t_k)) \approx \sum_{l_1=0}^{L} \sum_{l_2=0}^{L} (\Omega_{kl_1} \Omega^*_{kl_2})(\Upsilon_{jl_1} \Upsilon^*_{il_2}) \tag{11}$$

1.1 Choosing decompositions

1.1.1 The exponential decomposition

Define $A^k_{ij} = \exp(\Phi(\mathbf{r}_j, t_k) + \Phi^*(\mathbf{r}_i, t_k)) = g_{jk} g^*_{ik}$. Where we further defined $g_{jk} := \exp(\Phi(\mathbf{r}_j, t_k))$. We seek a approximation of A^k_{ij} of the form

$$\hat{A}^k_{ij} = \sum_{p=0}^{P} \Lambda_{kp} \Theta_{jp} \Theta^*_{ip} \tag{12}$$

We use the usual Frobenius norm and seek to minimize the loss

$$\mathcal{L} = \sum_k \|A^k - \hat{A}^k\|_F^2 \tag{13}$$

Using Wirtinger calculus, we can compute the gradient with respect to Θ_{jp} as

$$\frac{\partial \mathcal{L}}{\partial \Theta^*_{jp}} = \sum_{k,i} \left(\sum_{p'=0}^{P} \Lambda_{kp'} \Theta_{jp'} \Theta^*_{ip'} - A^k_{ij} \right) \Lambda^*_{kp} \Theta_{ip} \tag{14}$$

and the gradient with respect to Λ_{kp} as

$$\frac{\partial \mathcal{L}}{\partial \Lambda^*_{kp}} = \sum_{j,i} \left(\sum_{p'=0}^{P} \Lambda_{kp'} \Theta_{jp'} \Theta^*_{ip'} - A^k_{ij} \right) \Theta_{jp} \Theta^*_{ip} \tag{15}$$

If we have orthogonality constraints on Θ_{jp},

$$\sum_j \Theta^*_{jp} \Theta_{jq} = \delta_{pq}, \tag{16}$$

then the gradients simplify.

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial \Theta^*_{jp}} &= \sum_{k,i} \left(\sum_{p'=0}^{P} \Lambda_{kp'} \Theta_{jp'} \Theta^*_{ip'} - g_{jk} g^*_{ik} \right) \Lambda^*_{kp} \Theta_{ip} \\
&= \sum_k \sum_{p'=0}^{P} \Lambda_{kp'} \Lambda^*_{kp} \Theta_{jp'} \left(\sum_i \Theta^*_{ip'} \Theta_{ip} \right) - \sum_k g_{jk} \Lambda^*_{kp} \sum_i g^*_{ik} \Theta_{ip} \\
&= \sum_k \left[|\Lambda_{kp}|^2 \Theta_{jp} - g_{jk} \Lambda^*_{kp} \sum_i g^*_{ik} \Theta_{ip} \right]
\end{aligned} \tag{17}$$

and

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial \Lambda_{kp}^*} &= \sum_{j,i} \Big(\sum_{p'=0}^{P} \Lambda_{kp'} \Theta_{jp'} \Theta_{ip'}^* - g_{jk} g_{ik}^* \Big) \Theta_{jp}^* \Theta_{ip} \\
&= \sum_{p'=0}^{P} \Lambda_{kp'} \Big(\sum_{j,i} \Theta_{jp'} \Theta_{ip'}^* \Theta_{jp}^* \Theta_{ip} \Big) - \sum_{j,i} g_{jk} g_{ik}^* \Theta_{jp}^* \Theta_{ip} \\
&= \sum_{p'=0}^{P} \Lambda_{kp'} \Big(\sum_{j} \Theta_{jp'} \Theta_{jp}^* \Big) \Big(\sum_{i} \Theta_{ip'}^* \Theta_{ip} \Big) - \sum_{j,i} g_{jk} g_{ik}^* \Theta_{jp}^* \Theta_{ip} \\
&= \sum_{p'=0}^{P} \Lambda_{kp'} \delta_{pp'} - \sum_{j,i} g_{jk} g_{ik}^* \Theta_{jp}^* \Theta_{ip} \\
&= \Lambda_{kp} - (\sum_{j} g_{jk} \Theta_{jp}^*)(\sum_{i} g_{ik}^* \Theta_{ip}) \\
&= \Lambda_{kp} - |\sum_{i} g_{ik} \Theta_{ip}|^2
\end{aligned} \tag{18}$$

Thus the optimal Λ_{kp} for fixed Θ_{jp} is given by

$$\Lambda_{kp} = |\sum_{i} g_{ik} \Theta_{ip}|^2. \tag{19}$$

Thus we arrive at the following alternating optimization scheme:

1. Initialize Θ_{jp}. If not initialy orthogonal, orthogonalize.
2. Update $\Lambda_{kp} = |\sum_i g_{ik} \Theta_{ip}|^2$.
3. Update $\Theta_{jp} = \sum_k \Lambda_{kp} g_{jk} \sum_i g_{ik}^* \Theta_{ip}$, then orthogonalize.
4. Repeat steps 2 and 3 until convergence.

1.1.2 SVD-optimal initialisation of Θ

The alternating scheme above requires an initialisation for Θ_{jp}. We derive the choice of Θ_{jp} that is optimal for minimising the summed normal-operator error $\sum_{i,j} |N_{ij} - \hat{N}_{ij}|^2$, where

$$N_{ij} = \sum_{k} g_{ik}^* g_{jk}, \qquad \hat{N}_{ij} = \sum_{p} \mu_p \Theta_{ip}^* \Theta_{jp}, \qquad \mu_p = \sum_{k} \Lambda_{kp}. \tag{20}$$

Note the conjugate sits on the i-index in both N_{ij} and $\hat{N}_{ij}$, consistent with the approximation $g_{jk} \approx \sum_p \Theta_{jp}$ (forward model uses Θ_{jp} plain, not conjugated).

Taking the element-wise conjugate of both sides (which does not change $\sum_{i,j} |\cdot|^2$):

$$\sum_{i,j} |N_{ij} - \hat{N}_{ij}|^2 = \sum_{i,j} \left| \sum_{k} g_{ik} g_{jk}^* - \sum_{p} \mu_p \Theta_{ip} \Theta_{jp}^* \right|^2. \tag{21}$$

The right-hand side is the Frobenius loss for approximating the matrix $\tilde{N}_{ij} := \sum_k g_{ik} g_{jk}^*$ by the rank-P Hermitian form $\sum_p \mu_p \Theta_{ip} \Theta_{jp}^*$. By the Eckart–Young theorem, the minimiser over orthonormal $\{\Theta_{jp}\}$ is the P leading eigenvectors of $\tilde{N}_{ij}$.

Now observe that $\tilde{N}_{ij} = \sum_k g_{ik} g_{jk}^*$ is exactly the normal operator of the *conjugated* forward model g_{jk}^*. The forward approximation (10) gives

$$g_{jk} \approx \sum_{l} \Omega_{kl} \Upsilon_{jl}. \tag{22}$$

Conjugating: $g^*_{jk} \approx \sum_l \Omega^*_{kl} \Upsilon^*_{jl}$. The normal operator of this conjugated model is

$$\tilde{N}_{ij} = \sum_k g_{ik} g^*_{jk} \approx \sum_l |\sigma_l|^2 \, \Upsilon^*_{il} \Upsilon_{jl}, \tag{23}$$

where σ_l are the singular values coming from the Ω_{kl} orthogonality. The eigenvectors of $\tilde{N}_{ij}$ are therefore Υ_{jp}, the same spatial modes already appearing in the decomposition of g_{jk}.

Thus the **optimal initialisation** is

$$\Theta_{jp} = \Upsilon_{jp}, \qquad \mu_p = \sigma_p^2. \tag{24}$$

Substituting into the closed-form optimal Λ_{kp}:

$$\begin{aligned} \Lambda_{kp} &= \left| \sum_i g^*_{ik} \, \Theta_{ip} \right|^2 = \left| \sum_i g^*_{ik} \, \Upsilon_{ip} \right|^2 \\ &= \left| \sum_i \sum_l \Omega^*_{kl} \Upsilon^*_{il} \, \Upsilon_{ip} \right|^2 = \left| \Omega^*_{kp} \right|^2 = |\Omega_{kp}|^2, \end{aligned} \tag{25}$$

where the last step uses the orthonormality $\sum_i \Upsilon^*_{il} \Upsilon_{ip} = \delta_{lp}$. Thus

$$\Lambda_{kp} = |\Omega_{kp}|^2. \tag{26}$$

The per-k weight Λ_{kp} is the squared magnitude of the k-space coefficient for mode p, which determines the Toeplitz kernel $h_p = \mathbf{F}_{\text{adj}}(\Lambda_{\cdot p})$ for the fast normal-operator application.

In practice g_{jk} is not available in factored form; the optimal Θ_{jp} is obtained via a randomised eigendecomposition of $\tilde{N}_{ij} = \sum_k g_{ik} g^*_{jk}$ (or equivalently of N_{ij}, whose conjugate eigenvectors equal those of $\tilde{N}$), keeping the P eigenvectors with largest eigenvalues.

1.1.3 Decomposition of $\Omega_{kl_1} \Omega^*_{kl_2}$

For each k we have a $L \times L$ hermitian matrix, thus we can extract the $L(L+1)/2$ unique elements to form a $(L(L+1)/2) \times K$ matrix on which we can perform a SVD truncated to the P'th largest singular values and reform the $L \times L$ matrix to get

$$\Omega_{kl_1} \Omega^*_{kl_2} \approx \sum_{p=0}^{P} C_{l_1 l_2, p} \Lambda_{kp}. \tag{27}$$

For each p we have a $L \times L$ hermitian matrix, for which we can perform a eigen-decomposition into the rank R approximation as

$$C_{l_1 l_2, p} \approx \sum_{r=0}^{R} \lambda_{rp} \hat{C}_{l_1 rp} \hat{C}^*_{l_2 rp}. \tag{28}$$

With $R = 1$ in (27) this would correspond to a Canonical Polyadic Decomposition.

Inserting these approximations into the normal operator (11) we get

$$\exp(\Phi(\mathbf{r}_j, t_k) + \Phi^*(\mathbf{r}_i, t_k)) \approx \sum_{l_1=0}^{L} \sum_{l_2=0}^{L} \sum_{p=0}^{P} \sum_{r=0}^{R} \Lambda_{kp} \lambda_{rp} \hat{C}_{l_1 rp} \hat{C}^*_{l_2 rp} \Upsilon_{jl_1} \Upsilon^*_{il_2} \tag{29}$$

$$= \sum_{p=0}^{P} \sum_{r=0}^{R} \Lambda_{kp} \lambda_{rp} \left(\sum_{l_1=0}^{L} \hat{C}_{l_1 rp} \Upsilon_{jl_1} \right) \left(\sum_{l_2=0}^{L} \hat{C}_{l_2 rp} \Upsilon_{il_2} \right)^* \tag{30}$$

$$= \sum_{p=0}^{P} \sum_{r=0}^{R} \Lambda_{kp} \lambda_{rp} \Theta_{jrp} \Theta^*_{irp} \tag{31}$$

1.1.4 Decomposition of $\Upsilon_{jl_1}\Upsilon^*_{il_2}$

Instead of decomposing the temporal factors $\Omega_{kl_1}\Omega^*_{kl_2}$, we may instead seek a decomposition of the spatial factors

$$\Upsilon_{jl_1}\Upsilon^*_{il_2}. \tag{32}$$

The goal is to obtain an approximation of the form

$$\Upsilon_{jl_1}\Upsilon^*_{il_2} \approx \sum_{p=0}^{P} C_{l_1 l_2, p}\Theta_{jp}\Theta^*_{ip}, \tag{33}$$

which would reduce the number of Fourier transforms required for application of the normal operator from L^2 to P.

Direct optimization of (33) is infeasible for realistic problem sizes, since the tensors $\Upsilon_{jl_1}\Upsilon^*_{il_2}$ cannot be explicitly formed or stored. Instead, we construct the decomposition using only matrix-vector products.

We begin by viewing Υ_{jl} as a matrix

$$\boldsymbol{\Upsilon} \in \mathbb{C}^{N\times L}, \tag{34}$$

where N is the number of spatial voxels. We then form the Hermitian operator

$$S_{ji} = \sum_{l=0}^{L} \Upsilon_{jl}\Upsilon^*_{il}. \tag{35}$$

This operator is never explicitly assembled. Instead, products $\mathbf{y} = \mathbf{S}\mathbf{x}$ are evaluated as

$$z_l = \sum_i \Upsilon^*_{il} x_i, \tag{36}$$

$$y_j = \sum_l \Upsilon_{jl} z_l. \tag{37}$$

Thus, application of $\mathbf{S}$ requires only multiplication by $\boldsymbol{\Upsilon}$ and $\boldsymbol{\Upsilon}^H$.

Using a Krylov subspace eigensolver such as Lanczos or randomized SVD, we compute the leading P eigenvectors of $\mathbf{S}$,

$$\sum_i S_{ji}\Theta_{ip} = \sigma_p^2\Theta_{jp}, \tag{38}$$

with orthonormality

$$\sum_j \Theta^*_{jp}\Theta_{jq} = \delta_{pq}. \tag{39}$$

The vectors Θ_{jp} span the optimal rank-P subspace in the sense of the Eckart–Young theorem applied to $\boldsymbol{\Upsilon}$.

Projecting Υ_{jl} onto this basis gives

$$B_{lp} = \sum_j \Theta^*_{jp}\Upsilon_{jl}, \tag{40}$$

such that

$$\Upsilon_{jl} \approx \sum_{p=0}^{P} \Theta_{jp}B_{lp}. \tag{41}$$

Substituting into $\Upsilon_{jl_1}\Upsilon^*_{il_2}$ yields

$$\Upsilon_{jl_1}\Upsilon^*_{il_2} \approx \sum_{p=0}^{P}\sum_{q=0}^{P} \Theta_{jp}B_{l_1 p}B^*_{l_2 q}\Theta^*_{iq}. \tag{42}$$

The desired decomposition (33) is obtained by neglecting off-diagonal couplings $p \neq q$, giving

$$\Upsilon_{jl_1}\Upsilon^*_{il_2} \approx \sum_{p=0}^{P} B_{l_1p}B^*_{l_2p}\Theta_{jp}\Theta^*_{ip}. \tag{43}$$

We therefore identify

$$C_{l_1l_2,p} = B_{l_1p}B^*_{l_2p}. \tag{44}$$

The quality of the approximation depends on the magnitude of the discarded cross terms $B_{l_1p}B^*_{l_2q}$ for $p \neq q$. To reduce these couplings, we may further rotate the basis vectors within the retained subspace. Let

$$\hat{\Theta}_{jp} = \sum_{q=0}^{P} \Theta_{jq}U_{qp}, \tag{45}$$

where $\mathbf{U} \in \mathbb{C}^{P\times P}$ is unitary. The projected coefficients become

$$\hat{B}_{lp} = \sum_{q} B_{lq}U_{qp}. \tag{46}$$

The rotation matrix $\mathbf{U}$ may then be chosen to minimize the off-diagonal energy

$$\mathcal{E}(\mathbf{U}) = \sum_{l}\sum_{p\neq q} |\hat{B}_{lp}\hat{B}^*_{lq}|^2. \tag{47}$$

Since this optimization is performed only over the small $P \times P$ unitary matrix $\mathbf{U}$, it remains computationally feasible even for large imaging problems. The optimization may be performed using Jacobi rotations, Riemannian gradient descent on the unitary group, or approximate joint diagonalization algorithms.

1.2 Feature vector and the histogram's scaling problem

The off-resonance and off-linear gradient terms only affect the signal on the support of the image, so the approximation in (10) only ever needs to run on voxels within the support. Consider the feature vector

$$\overline{\Psi}(\mathbf{r}) = \begin{bmatrix} \mathbf{z}(\mathbf{r}) = \Delta B_0(\mathbf{r}) + 1/T_2(\mathbf{r}) \\ R_0^0(\mathbf{r}) \\ R_1^0(\mathbf{r}) \\ R_1^{-1}(\mathbf{r}) \\ \vdots \end{bmatrix}. \tag{48}$$

The original implementation *binned* this feature vector by value (one histogram axis per component) and ran the SVD of (10) on the histogram bins instead of on individual voxels. This has a fatal scaling problem: if the feature vector has D_f components and component d is binned into n_d values, the number of histogram cells is

$$N_{\text{hist}} = \prod_{d=1}^{D_f} n_d, \tag{49}$$

independent of how many voxels are actually present — a product over bin *resolutions*, not a count of populated cells. With a non-trivial T_2 map and several concomitant channels active, N_{hist} routinely exceeds N, the number of support voxels, which defeats the entire point of histogramming (meant to *reduce* the problem below N before the SVD, not multiply it beyond N). Fixed bin-edge quantization is also itself a source of error the SVD step cannot see or correct for. **Conclusion: histogramming is removed entirely.** Every support voxel keeps its own exact feature value $\overline{\Psi}(\mathbf{r}_j)$ — the trivial upper bound on both accuracy (zero quantization error) and the number of rows the SVD ever sees (N, never more).

1.3 Why this doesn't make the SVD worse

Removing the histogram does not change what is being approximated, only how many (and which) rows go into the SVD of (10) — every row is now an exact voxel rather than a quantized bin centroid. By the Eckart–Young theorem (already invoked in Section 1.1.2 for the normal-operator initialisation), the SVD-truncated rank-L approximation is the *provably optimal* rank-L approximation of whatever matrix it is given, in Frobenius norm — no fixed, non-data-adapted basis (Chebyshev, polynomial, or otherwise) can do better for the same L. So the right fix for the histogram's blowup is not to replace the SVD with something cheaper-but-suboptimal; it is to give the SVD the exact matrix directly (matrix-free matvecs over the real voxels, $O(N)$ per matvec) instead of a value-quantized, combinatorially-sized surrogate of it.

1.4 An exact nonlinear (multiplicative) decomposition

The bilinear structure of Φ_{res} suggests a genuinely different lever, orthogonal to the SVD discussion above: since $\Delta B_0(\mathbf{r})$, $1/T_2(\mathbf{r})$, and $N_q(\mathbf{r})$ are all static (time-independent), the residual exponent of (10) is *exactly linear* in the feature vector $\overline{\Psi}(\mathbf{r})$ for fixed t:

$$\Phi_{\text{res}}(\mathbf{r}, t) = -t\, z(\mathbf{r}) + \sum_{q=1}^{Q_c} N_q(\mathbf{r})\, W_q(t) = \overline{\Psi}(\mathbf{r}) \cdot \mathbf{w}(t), \tag{50}$$

where $W_q(t)$ is channel q's known time-integral — for the static-GNL channels already absorbed into the warp this is a single gradient integral $\int_0^t G_\beta(\tau)\, d\tau$, but for *concomitant* channels it is the gradient-*product* integral demanded by the 2nd-order Maxwell terms (e.g. $W_{z^2}(t) = -i\gamma \int_0^t (G_x(\tau)^2 + G_y(\tau)^2)/(8B_0)\, d\tau$). Separability only requires that each term factor into a pure function of $\mathbf{r}$ times a pure function of t — it does not require W_q to be linear in any single gradient waveform, so the gradient-product origin of the concomitant terms does not break (50).

This bilinear exponent has an **exact** multiplicative factorization, with zero approximation error:

$$g(\mathbf{r}, t) = \exp\big(\overline{\Psi}(\mathbf{r}) \cdot \mathbf{w}(t)\big) = \prod_{d=1}^{D_f} \exp\big(\Psi_d(\mathbf{r})\, w_d(t)\big), \tag{51}$$

i.e. D_f factors, one per physical channel, each *exactly* rank-1 in its own axis. This is a genuinely nonlinear (multiplicative, not additive) identity — $\exp(A)\exp(B) = \exp(A+B)$ run in reverse — and is not a Taylor truncation: it is exact at every order, regardless of how large $\Psi_d(\mathbf{r}) w_d(t)$ gets.

Why this doesn't trivially collapse L. The NUFFT/Toeplitz-embedding machinery above only computes a single linear sum over $\mathbf{r}$ for one t at a time. Multiplying D_f separate t-dependent factors into the image before gridding still requires, in general, a fresh per-sample multiply for every one of the K readout times, unless *each factor on its own* already has low rank in t. It does: this is exactly the classical 1D time-segmentation result — $\exp(z(\mathbf{r})t)$ is smooth and bounded over a finite readout, so it needs only a handful of terms. The same admissibility argument applies per concomitant channel independently.

The actual lever: composition instead of fusion. Decompose each channel *separately* in time — its own small L_d, fit by the same per-channel closed-form weighted least squares used for classical 1D time-segmentation,

$$\mathbf{\Omega}^{(d)}_{k,:} = \Big(\mathbf{B}_d^H \mathbf{D}_w \mathbf{B}_d\Big)^{-1} \mathbf{B}_d^H \mathbf{D}_w\, \exp\big(\Psi_d(\cdot)\, w_d(t_k)\big), \tag{52}$$

where $\mathbf{B}_d$ is any small fixed basis (e.g. Chebyshev/Lagrange) on axis d alone — applied one axis at a time — and then composed as **sequential** Toeplitz/NUFFT stages within each CG/gradient-step iteration, rather than expanding (51) into one fused additive sum (which reproduces the *same* $\prod_d L_d$ blow-up as the histogram, just relocated from bin count to term count). Composing factors at *application time* instead of fusing them into one basis costs

$$\sum_{d=1}^{D_f} L_d \quad \text{(composition)} \qquad \text{vs.} \qquad \prod_{d=1}^{D_f} L_d \quad \text{(fused sum)}, \tag{53}$$

i.e. additive instead of multiplicative scaling in the number of channels — a real, exact nonlinear-decomposition win, distinct from (and compatible with) the per-channel SVD/time-segmentation choice of each L_d. The price is more (but individually cheaper) NUFFT/Toeplitz applications per iteration instead of one larger fused one, and the requirement that the iterative solver actually apply the channels as a chain rather than assuming a single fused forward operator.

A literal elementwise nonlinearity inside one flat sum (e.g. fitting $g \approx \big(\sum_l a_l(\mathbf{r})b_l(t)\big)^2$ or $\sum_l \sin(\cdot)$ terms) does *not* have this clean structure: the NUFFT can only apply a *linear* sum of complex exponentials efficiently, so feeding its output through a further elementwise nonlinearity breaks the "multiply image, then NUFFT" pattern that makes (10)–(73) fast in the first place. The exponential's own multiplicativity (51), realised as *operator composition*, is the form of nonlinearity that is actually exploitable here.

Normal Operator via Toeplitz Embedding

Given the approximation (10), the discretised forward operator for coil c is

$$(\mathbf{A}_c\boldsymbol{\rho})_k = \sum_l \Omega_{kl}\big[\mathbf{F}\,\mathrm{diag}(\Upsilon_l \odot s_c)\,\boldsymbol{\rho}\big]_k, \tag{54}$$

where $\mathbf{F}$ is the NUFFT and $(\Upsilon_l)_j = \Upsilon_{jl}$. The adjoint is

$$(\mathbf{A}_c^H\mathbf{s})_j = s_c^*(\mathbf{r}_j)\sum_l \Upsilon_{jl}^*\,\big[\mathbf{F}^H\,\mathrm{diag}(\Omega_l^*)\,\mathbf{s}\big]_j. \tag{55}$$

Toeplitz structure

Define the PSF for k-space weights $w \in \mathbb{C}^K$ as the type-1 NUFFT

$$h_w = \mathbf{F}_{\mathrm{adj}}(w), \qquad (h_w)_n = \sum_k w_k\, e^{i\mathbf{k}(t_k)\cdot\mathbf{r}_n}. \tag{56}$$

The Toeplitz application uses a zero-padded FFT of size $2N$ in each dimension:

$$\big[\mathbf{F}^H\,\mathrm{diag}(w)\,\mathbf{F}\,f\big] = h_w \star f = \mathrm{IFFT}\big[\hat{H}_w \odot \mathrm{FFT}(f_{\mathrm{pad}})\big], \tag{57}$$

where $\hat{H}_w = \mathrm{FFT}(h_{w,\mathrm{pad}})$ is precomputed once.

L^2 Toeplitz approach

Using (11) directly:

$$(\mathbf{A}_c^H\mathbf{A}_c\,\boldsymbol{\rho})_i \approx \sum_{l_1=1}^{L}\sum_{l_2=1}^{L}\Upsilon_{il_2}^*\,\big(h_{l_1l_2}\star(\Upsilon_{l_1}\odot s_c\odot\boldsymbol{\rho})\big)_i, \tag{58}$$

with one kernel per pair:

$$h_{l_1l_2} = \mathbf{F}_{\mathrm{adj}}(\Omega_{l_1}\odot\Omega_{l_2}^*). \tag{59}$$

Setup cost: L^2 type-1 NUFFTs. Application cost per iteration: L^2 Toeplitz multiplications (57).

$P \times R$ Toeplitz approach (reduced)

Inserting (31):

$$(\mathbf{A}_c^H\mathbf{A}_c\,\boldsymbol{\rho})_i \approx \sum_{p=1}^{P}\sum_{r=1}^{R}\Theta_{irp}^*\,\big(h_p\star(\Theta_{rp}\odot\boldsymbol{\rho})\big)_i, \tag{60}$$

where P distinct kernels (one per p):

$$h_p = \mathbf{F}_{\mathrm{adj}}(\Lambda_p), \tag{61}$$

and the combined spatial weights:

$$\Theta_{jrp} = s_c(\mathbf{r}_j) \sum_{l=1}^{L} \hat{C}_{lrp}\, \Upsilon_{jl}. \tag{62}$$

Setup cost: P type-1 NUFFTs (independent of L). Application cost per iteration: $P \times R$ Toeplitz multiplications. With $R = 1$ this is P multiplications, and if $P \ll L^2$ we achieve a large speedup.

Algorithm (setup)

1. Compute $\Omega \in \mathbb{C}^{K\times L}$, $\Upsilon \in \mathbb{C}^{N_{\text{hist}}\times L}$ from SVD of e^{Φ}.
2. Form the $\frac{L(L+1)}{2} \times K$ matrix of unique upper-triangular products $\Omega_{kl_1}\Omega^*_{kl_2}$.
3. Truncated SVD of rank $P \Rightarrow \mathbf{\Lambda} \in \mathbb{C}^{K\times P}$, coefficient matrices $\mathbf{C}_p \in \mathbb{C}^{L\times L}$.
4. Rank-R SVD of each $\mathbf{C}_p \Rightarrow \hat{C}_{lrp}$.
5. $\Theta_{jrp} = \sum_l \hat{C}_{lrp}\, \Upsilon_{jl}$ scattered to full voxel grid.
6. P Toeplitz kernels: $\hat{H}_p = \text{FFT}\big[\mathbf{F}_{\text{adj}}(\Lambda_p)_{\text{pad}}\big]$.

Normal operator application

$$(\mathbf{A}^H\mathbf{A}\,\boldsymbol{\rho})_i = \sum_{p=1}^{P}\sum_{r=1}^{R} \Theta^*_{irp}\, \text{IFFT}\big[\hat{H}_p \odot \text{FFT}((\Theta_{rp}\odot\boldsymbol{\rho})_{\text{pad}})\big]_i. \tag{63}$$

2 Non-Hermitian Decomposition and Multi-Coil Normal Operator

2.1 Non-Hermitian Approximation

The hermitian decomposition of the normal-operator phase factor

$$e^{\Phi(\mathbf{r}_j,t_k)+\Phi^*(\mathbf{r}_i,t_k)} \approx \sum_{p=0}^{P} \Lambda_{kp}\, \Theta_{jp}\Theta^*_{ip} \tag{64}$$

constrains the same spatial basis Θ to appear on both the j- and i-index, and forces $\Lambda_{kp} \geq 0$. We generalise by allowing independent spatial functions for each index:

$$e^{\Phi(\mathbf{r}_j,t_k)+\Phi^*(\mathbf{r}_i,t_k)} \approx \sum_{p=0}^{P} \Lambda_{kp}\, A_{jp}\, B_{ip}, \tag{65}$$

where A_{jp}, $B_{ip} \in \mathbb{C}$ are independent spatial basis functions and $\Lambda_{kp} \in \mathbb{C}$. Here B is defined so that the hermitian case corresponds to $B_{ip} = \Theta^*_{ip}$, $A_{jp} = \Theta_{jp}$, $\Lambda_{kp} \in \mathbb{R}_{\geq 0}$. The more general form (65) strictly enlarges the approximation class and can achieve smaller error for the same number of terms P.

2.2 Full Normal Operator with Coil Maps and Coil Covariance

We consider C receive coils with sensitivity maps $s_c(\mathbf{r})$ and noise covariance matrix $\Psi \in \mathbb{C}^{C\times C}$ (positive definite Hermitian). The per-coil forward operator element is

$$(A_c)_{kj} = s_{cj}\, e^{\Phi(\mathbf{r}_j,t_k)}\, e^{-i\mathbf{k}(t_k)\cdot\mathbf{r}_j}, \tag{66}$$

and the maximum-likelihood (whitened) normal operator is

$$N_{ij} = \sum_{c,c'} (\Psi^{-1})_{cc'} \sum_k (A_c^H)_{ik}(A_{c'})_{kj} = \sum_{c,c'} (\Psi^{-1})_{cc'}\, s^*_{ci}\, s_{c'j} \sum_k e^{\Phi^*(\mathbf{r}_i,t_k)+\Phi(\mathbf{r}_j,t_k)} e^{-i\mathbf{k}(t_k)\cdot(\mathbf{r}_j-\mathbf{r}_i)}. \tag{67}$$

Define the coil correlation factor

$$\Xi_{ij} = \mathbf{s}_i^H \, \Psi^{-1} \, \mathbf{s}_j = \sum_{c,c'} (\Psi^{-1})_{cc'} \, s_{ci}^* \, s_{c'j}, \quad [\mathbf{s}_j]_c = s_{cj}. \tag{68}$$

Inserting the non-hermitian approximation (65) and defining the Toeplitz kernels

$$h_p(\mathbf{r}) = \sum_k \Lambda_{kp} \, e^{-i\mathbf{k}(t_k)\cdot\mathbf{r}} = \big[\mathbf{F}_{\mathrm{adj}}(\Lambda_{\cdot p})\big](\mathbf{r}), \tag{69}$$

the full normal operator element becomes

$$N_{ij} \approx \big(\mathbf{s}_i^H \, \Psi^{-1} \, \mathbf{s}_j\big) \sum_p A_{jp} \, B_{ip} \; h_p(\mathbf{r}_j - \mathbf{r}_i). \tag{70}$$

The Toeplitz convolution structure is preserved because h_p depends only on the difference $\mathbf{r}_j - \mathbf{r}_i$. Note that the kernels h_p are complex-valued when $\Lambda_{kp} \in \mathbb{C}$.

2.3 Operator Application and Whitened Coil Maps

The action of (70) on a vector $\boldsymbol{\rho}$ is

$$(N\boldsymbol{\rho})_i \approx \sum_p B_{ip} \sum_{c,c'} (\Psi^{-1})_{cc'} \, s_{ci}^* \Big[h_p \star \big(s_{c'} \odot A_p \odot \boldsymbol{\rho} \big) \Big]_i. \tag{71}$$

To expose the structure more cleanly, factor $\Psi^{-1} = L^H L$ (e.g. via Cholesky) and define the *whitened coil maps*

$$\tilde{s}_{qj} = \big[L \, \mathbf{s}_j \big]_q, \qquad q = 1, \ldots, C. \tag{72}$$

Then $\Xi_{ij} = \sum_q \tilde{s}_{qi}^* \, \tilde{s}_{qj}$ and (71) becomes

$$\boxed{(N\boldsymbol{\rho})_i \approx \sum_p B_{ip} \sum_q \tilde{s}_{qi}^* \Big[h_p \star \big(\tilde{s}_q \odot A_p \odot \boldsymbol{\rho} \big) \Big]_i.} \tag{73}$$

The coil covariance enters only through the whitened maps $\tilde{s}_q$. The per-iteration cost is $P \times C$ Toeplitz multiplications, identical in structure to the hermitian case but with the flexibility of using different input (A_p) and output (B_p) spatial modes. If Ψ^{-1} is well-approximated by a rank-Q matrix ($Q < C$), the inner sum truncates to Q virtual coils.

Per-mode algorithm

For each mode $p = 1, \ldots, P$:

1. For each whitened coil q: form $v_j^{(q)} = \tilde{s}_{qj} \, A_{jp} \, \rho_j$.
2. For each q: Toeplitz-convolve $u_i^{(q)} = \big[h_p \star v^{(q)} \big]_i$.
3. Accumulate: $(N\boldsymbol{\rho})_i \mathrel{+}= B_{ip} \sum_q \tilde{s}_{qi}^* \, u_i^{(q)}$.

2.4 Optimal Decomposition: Alternating Optimization

Scale and implicit operators

In practice $N, K \sim 10^5$–10^6, so the matrix $G \in \mathbb{C}^{N \times K}$ with $G_{jk} = g_{jk} = e^{\Phi(\mathbf{r}_j, t_k)} \, e^{-i\mathbf{k}(t_k)\cdot\mathbf{r}_j}$ is never formed or stored. All computations below involve only matrix-vector products of the form

$$[G \, \mathbf{v}]_j = \sum_k g_{jk} \, v_k, \qquad [G^T \mathbf{x}]_k = \sum_j g_{jk} \, x_j, \qquad [\bar{G} \, \mathbf{v}]_j = \sum_k g_{jk}^* \, v_k, \qquad [\bar{G}^T \mathbf{x}]_k = \sum_j g_{jk}^* \, x_j, \tag{74}$$

each of which is one (possibly conjugated) NUFFT application of cost $O(N \log N + K)$. The SVD needed for initialisation is computed on a small histogram-reduced matrix $G_{\mathrm{hist}} \in \mathbb{C}^{N_{\mathrm{hist}} \times K}$ with $N_{\mathrm{hist}} \ll N$ as described in Section 1.2.

Loss and constraints

We seek $A \in \mathbb{C}^{N\times P}$, $B \in \mathbb{C}^{N\times P}$, $\Lambda \in \mathbb{C}^{K\times P}$ minimising

$$\mathcal{L} = \sum_{i,j,k} \Big| g_{jk} g_{ik}^* - \sum_{p=0}^{P} \Lambda_{kp} A_{jp} B_{ip} \Big|^2, \tag{75}$$

under the orthonormality constraints $A^H A = I_P$ and $B^H B = I_P$.

Update for Λ (fixed A, B)

Define the projected temporal matrices

$$\mathbf{F} = G^T \overline{A} \in \mathbb{C}^{K\times P}, \quad F_{kp} = \sum_j g_{jk}\, \overline{A_{jp}}, \qquad \mathbf{C} = \overline{G}^T \overline{B} \in \mathbb{C}^{K\times P}, \quad C_{kp} = \sum_j g_{jk}^*\, \overline{B_{jp}}. \tag{76}$$

Each column of $\mathbf{F}$ and $\mathbf{C}$ costs one NUFFT. Setting $\partial\mathcal{L}/\partial\Lambda_{kp}^* = 0$ under $A^H A = I$, $B^H B = I$ gives

$$\mathbf{\Lambda} = \mathbf{F} \odot \mathbf{C}. \tag{77}$$

In the hermitian case ($B_{ip} = \Theta_{ip}^*$, $A_{jp} = \Theta_{jp}$) this reduces to $\Lambda_{kp} = |\sum_j g_{jk} \overline{\Theta_{jp}}|^2 \geq 0$. In general $\Lambda_{kp} \in \mathbb{C}$.

Update for A (fixed B, Λ)

Setting $\partial\mathcal{L}/\partial\overline{A_{jp}} = 0$ under $B^H B = I$ gives

$$A_{\cdot p} \leftarrow G\big(\overline{\Lambda_{\cdot p}} \odot \mathbf{C}_{\cdot p}\big), \tag{78}$$

i.e. one G-matvec per mode, followed by orthogonalisation of A.

Update for B (fixed A, Λ)

Setting $\partial\mathcal{L}/\partial\overline{B_{ip}} = 0$ under $A^H A = I$ gives

$$B_{\cdot p} \leftarrow \bar{G}\big(\overline{\Lambda_{\cdot p}} \odot \mathbf{F}_{\cdot p}\big), \tag{79}$$

i.e. one $\bar{G}$-matvec per mode (conjugate NUFFT), followed by orthogonalisation of B. Note the symmetry: both A and B updates use $\overline{\Lambda_p}$, but A uses G and B uses $\bar{G}$.

Cost per ALS iteration

Each pass through updates (77)–(79) requires $4P$ NUFFT applications (2 for $\mathbf{F}, \mathbf{C}$; P each for the A and B updates) plus two $O(NP^2)$ Gram–Schmidt orthogonalisations, cheap for small P.

Algorithm

1. Compute SVD of $G_{\text{hist}} = U_{\text{hist}} \Sigma V^H$. Select top-$P$ pairs (l_1, l_2) by $\sigma_{l_1}\sigma_{l_2}$ (see below). Set $A_{\cdot p} = (U_{\text{hist}})_{\cdot l_1(p)}$, $B_{\cdot p} = \overline{(U_{\text{hist}})_{\cdot l_2(p)}}$, $\Lambda_{kp} = \sigma_{l_1(p)}\sigma_{l_2(p)} V^*_{k,l_1(p)} V_{k,l_2(p)}$.
2. Orthogonalise A and B via QR; recompute $\mathbf{\Lambda} = \mathbf{F} \odot \mathbf{C}$.
3. Compute $\mathbf{F} = G^T \overline{A}$ and $\mathbf{C} = \overline{G}^T \overline{B}$.
4. Update $\mathbf{\Lambda} = \mathbf{F} \odot \mathbf{C}$.
5. Update $A_{\cdot p} \leftarrow G(\overline{\Lambda_{\cdot p}} \odot \mathbf{C}_{\cdot p})$ for each p; orthogonalise A.
6. Update $B_{\cdot p} \leftarrow \bar{G}(\overline{\Lambda_{\cdot p}} \odot \mathbf{F}_{\cdot p})$ for each p; orthogonalise B.
7. Repeat steps 3–6 until $\mathcal{L}$ converges.

Optimal SVD initialisation

The full L^2 expansion of the target reads

$$g_{jk}g_{ik}^* = \sum_{l_1=1}^{L}\sum_{l_2=1}^{L} \sigma_{l_1}\sigma_{l_2}\, U_{jl_1}U_{il_2}^*\, V_{kl_1}^* V_{kl_2}. \tag{80}$$

Each pair (l_1, l_2) contributes a term of magnitude $\sigma_{l_1}\sigma_{l_2}$. In our convention $B_{ip} = U_{il_2(p)}^*$ (absorbing the conjugate into B), the pair-p contribution is $\Lambda_{kp}A_{jp}B_{ip} = \sigma_{l_1}\sigma_{l_2}U_{jl_1}U_{il_2}^*V_{kl_1}^*V_{kl_2}$. Retaining only diagonal pairs (l, l) recovers the hermitian case $B_{ip} = \Theta_{ip}^* = U_{ip}^*$, $\Lambda_{kp} = \sigma_p^2|V_{kp}|^2 = |\Omega_{kp}|^2$.

For the non-hermitian initialisation, rank all L^2 pairs by $\sigma_{l_1}\sigma_{l_2}$ and retain the top P. The off-diagonal pair $(1, 2)$ contributes $\sigma_1\sigma_2$, which exceeds the diagonal contribution σ_P^2 of the last hermitian term whenever $\sigma_P < \sqrt{\sigma_1\sigma_2}$ — almost always true for a fast-decaying spectrum. Since repeated indices occur (e.g. $U_{\cdot 1}$ appears in pairs $(1,1), (1,2), (1,3), \ldots$), the A and B columns are not orthonormal as initialised; QR orthogonalisation followed by a $\mathbf{\Lambda}$ recompute (step 2 of the algorithm) restores the constraint.

2.5 Comparison Summary

	Hermitian	Non-Hermitian
Decomposition	$\sum_p \Lambda_{kp}\Theta_{jp}\Theta_{ip}^*$	$\sum_p \Lambda_{kp}A_{jp}B_{ip}$
Λ_{kp}	real, ≥ 0	complex
Input spatial weight	$\Theta_p \odot \tilde{s}_q$	$A_p \odot \tilde{s}_q$
Output spatial weight	$\Theta_p^* \odot \tilde{s}_q^*$	$B_p \odot \tilde{s}_q^*$
Toeplitz kernel	$\mathbf{F}_{\text{adj}}(\Lambda_{\cdot p})$, real Λ	same, complex Λ
Cost per iteration	$P \times C$ Toeplitz	$P \times C$ Toeplitz

3 Exact Closed-Form Expansion for Concomitant Fields

Off-resonance/T_2 ($z(\mathbf{r})$) and concomitant fields were both routed through the same generic data-driven machinery (histogram-then-SVD, or per-voxel SVD after Section 1.2) in earlier sections. That machinery is the right tool for $z(\mathbf{r})$, a genuinely unknown, patient-specific field with no closed form. It is the *wrong* tool for concomitant fields, which are entirely determined by sequence design before any data is acquired — using a generic, data-fitted basis for something already known in closed form throws away exact structure for no reason, and is also why naive treatments (low-order polynomial phase fits applied to the *reconstructed image*) fail: they fit the wrong domain. Concomitant phase varies with the acquisition timeline (per-sample t_k, with a slope that changes every spoke), not as a single static spatial pattern superimposed on the final image — a post-hoc fit on the image can only remove a t-independent phase, which this is not. The fix is to put the exact structure into the operator itself, at the point where t_k is still present.

3.1 The spatial basis is already minimal

The standard 2nd-order Maxwell concomitant field (Bernstein/King/Zhou; Norris & Hutchison 1990) is, for a z-main-field system with instantaneous gradient $\mathbf{G}(t) = (G_x, G_y, G_z)(t)$,

$$B_{\text{conc}}(\mathbf{r}, t) = \frac{G_x^2 + G_y^2}{8B_0}z^2 + \frac{G_z^2}{8B_0}(x^2 + y^2) + \frac{G_xG_z}{2B_0}xz + \frac{G_yG_z}{2B_0}yz. \tag{81}$$

Expanding in the 6 monomials of $\mathbf{G}(t)$ — $\{G_x^2, G_y^2, G_z^2, G_xG_y, G_xG_z, G_yG_z\}$ — the G_x^2 and G_y^2 terms share the identical spatial weight $z^2/(8B_0)$, and the G_xG_y coefficient is exactly zero (there is no xy term in the standard theory). What remains is exactly $Q_c = 4$ linearly independent spatial fields, $\{z^2, x^2 + y^2, xz, yz\}$. This is the *minimal* spatial rank — it follows directly from (81), not from fitting or truncating anything, and cannot be reduced further (the four fields are linearly independent functions of $\mathbf{r}$).

3.2 The temporal coefficients are exactly known

Define $\Phi_{\text{conc}}(\mathbf{r},t) = \sum_{q=1}^{Q_c} N_q(\mathbf{r})\, w_q(t)$, where $N_q \in \{z^2, x^2+y^2, xz, yz\}$ and

$$w_q(t) = -i\gamma \int_0^t \text{coupling}_q(\tau)\, d\tau, \tag{82}$$

with $\text{coupling}_q(t)$ the corresponding gradient-product factor from (81) (e.g. $(G_x(t)^2 + G_y(t)^2)/(8B_0)$ for q=1). Unlike $z(\mathbf{r})$, $\text{coupling}_q(t)$ depends only on the pulse sequence's own gradient waveforms — it requires no field map, no calibration scan, no fitting of any kind. $w_q(t)$ is computable to machine precision directly from the sequence design.

3.3 Low-order Taylor expansion, with an a priori error bound

Since both the spatial basis $\{N_q\}$ and the temporal coefficients $w_q(t)$ are exactly known, the multinomial (Taylor) expansion of $\exp(\Phi_{\text{conc}})$ is itself exactly known, with no data-dependent step anywhere:

$$\exp\big(\Phi_{\text{conc}}(\mathbf{r},t)\big) = \sum_{p=0}^{\infty} \frac{1}{p!}\Big(\sum_{q=1}^{Q_c} N_q(\mathbf{r}) w_q(t)\Big)^p = \sum_{\alpha} \frac{1}{\alpha!}\, F_\alpha(\mathbf{r})\, g_\alpha(t), \tag{83}$$

where $\alpha = (\alpha_1, \ldots, \alpha_{Q_c})$ ranges over multi-indices, $\alpha! = \prod_q \alpha_q!$, $F_\alpha(\mathbf{r}) = \prod_q N_q(\mathbf{r})^{\alpha_q}$ (a fixed, exactly known polynomial in $\mathbf{r}$ — e.g. $\alpha = (2,0,0,0)$ gives $F_\alpha = z^4$), and $g_\alpha(t) = \prod_q w_q(t)^{\alpha_q}$ (a fixed, exactly known function of t, computable directly from the sequence). Truncating at total order $|\alpha| \le P$ gives

$$\exp\big(\Phi_{\text{conc}}(\mathbf{r},t)\big) \approx \sum_{|\alpha|\le P} \frac{1}{\alpha!}\, F_\alpha(\mathbf{r})\, g_\alpha(t), \qquad L(P) = \binom{Q_c + P}{P} \tag{84}$$

terms — $L(1) = 5$ (the constant term plus exactly the existing 4 channels), $L(2) = 15$, and so on. **Every coefficient in** (84) **is exact and known in closed form** — there is no SVD, no histogram, no fitting step for the concomitant block at any truncation order.

The truncation error is the standard exponential Taylor remainder,

$$\Big|\exp(\Phi_{\text{conc}}) - \sum_{|\alpha|\le P} \tfrac{1}{\alpha!} F_\alpha g_\alpha\Big| \;\le\; \frac{|\Phi_{\text{conc}}|^{P+1}}{(P+1)!}\, e^{|\Phi_{\text{conc}}|}, \tag{85}$$

giving a concrete, checkable a priori bound from $\max_{\mathbf{r},t} |\Phi_{\text{conc}}|$ alone — exactly the quantity already printed as the "concomitant: max phase" diagnostic. This is why low order is justified here specifically: concomitant phase scales as G^2/B_0 (quadratic in gradient amplitude over twice the main field), typically a small number of radians at most for realistic sequences/field strengths, unlike the off-resonance or static-GNL phases, which can range over many radians and therefore cannot use a low-order Taylor truncation (they need the genuine admissibility/Chebyshev compression of Section 1.2's warp discussion, or real data-driven SVD). *This is the precise sense in which "fit a low-order polynomial" intuition is correct for concomitant fields but wrong when applied post-hoc to the image*: the expansion converges quickly in the right variable (Φ_{conc}, evaluated at acquisition time t_k), not in image-domain phase after the fact.

3.4 What this does *not* buy: the normal operator

(84) has exactly the $\sum_l \Omega_{kl}\Upsilon_{jl}$ form of (10) for the *forward* operator — but the forward operator is applied once (to form $\mathbf{A}^H\mathbf{b}$); every iteration's cost is the *normal* operator $\mathbf{A}^H\mathbf{A}$, and a cheap forward decomposition does not imply a cheap normal operator. Composing this exact concomitant block with the off-resonance block sequentially, as originally suggested here, is wrong: the true exponent is a *sum* $\Phi_{\text{total}} = -tz(\mathbf{r}) + \Phi_{\text{conc}}(\mathbf{r},t)$, so $\exp(\Phi_{\text{total}}) = \exp(-tz(\mathbf{r}))\exp(\Phi_{\text{conc}})$ is a *product*, not a sum, of two factors multiplying the same image at the same acquisition time — there is no stage at which they act on separate inputs, so there is

nothing to compose sequentially. Combining an L_z-term off-resonance decomposition with an $L_{\text{conc}} = L(P)$-term concomitant decomposition gives $L_z \cdot L_{\text{conc}}$ forward terms, full stop, and Taylor's exactness for the concomitant factor does not change that.

Worse, even granting the $L_z \cdot L_{\text{conc}}$ forward terms, this is feeding the *normal*-operator compression of Section 4 below a deliberately non-data-adapted (Taylor, not SVD) intermediate basis. Since SVD is optimal at every stage (Eckart–Young), chaining "Taylor → compress" is never better than "direct per-voxel SVD on the combined $\exp(\Phi_{\text{total}}) \to$ compress," and is generally worse for the same final rank P. **Conclusion: build (Ω, Υ) once, directly, via the matrix-free per-voxel SVD of Section 1.3 on the combined off-resonance+concomitant exponent, and feed that into the normal-operator compression below.** The exact concomitant structure derived above is not useless — (85)'s a priori error bound is a legitimate, free diagnostic for how many SVD modes the concomitant contribution is likely to need, and the closed-form $g_\alpha(t)$ is a cheap way to evaluate matvecs during that SVD — but it is a construction-cost convenience, not a rank reduction.

3.5 Reducing the per-term cost of the $L_z \times Q_c$ coupling, via k-space differentiation

The $L_z \cdot Q_c$ term count above is real and unavoidable for a per-term correction — it is not removed by anything below. What *can* be reduced is the cost of realizing each of those $L_z \cdot Q_c$ terms. The standard Fourier property

$$\mathcal{F}\big[r_a r_b\, f(\mathbf{r})\big](\mathbf{k}) = -\partial_{k_a}\partial_{k_b}\,\hat{f}(\mathbf{k}), \qquad \hat{f} = \mathcal{F}[f], \tag{86}$$

is exact for any axis pair a, b (including $a{=}b$) and any function f. Since every N_q is a quadratic monomial (Section 3 above), $\mathcal{F}[N_q \cdot f]$ is exactly $D_q[\hat{f}]$ for a fixed second-derivative operator $D_q \in \{-\partial_{k_z}^2,\ -(\partial_{k_x}^2 + \partial_{k_y}^2),\ -\partial_{k_x}\partial_{k_z},\ -\partial_{k_y}\partial_{k_z}\}$.

The existing L_z-term normal operator already computes $\widehat{\Upsilon_l \rho} = \mathcal{F}[\Upsilon_l \cdot \rho]$ once per term l, as part of the unmodified base Toeplitz machinery. Rather than building a *new* NUFFT/kernel for each (Υ_l, N_q) pair (the $L_z \cdot Q_c$ forward terms of the previous subsection), apply D_q directly to the already-computed array:

$$\mathcal{F}\big[N_q \cdot \Upsilon_l \cdot \rho\big] = D_q\big[\widehat{\Upsilon_l \rho}\big], \tag{87}$$

evaluated via a finite-difference stencil on the same k-space grid the base term already populates. D_q (a second derivative) is self-adjoint under the periodic boundary conditions the FFT grid already has, so the same trick applies symmetrically in the adjoint/gridding direction — the correction's adjoint is D_q applied *before* the existing inverse FFT, not after, with no extra transform either way.

This does not change the $L_z \cdot Q_c$ term count, and is not free: it trades exactness for a small-Φ_{conc} truncation (only the $P{=}1$ term of (84) is reachable this way without combinatorially many higher-derivative stencils — consistent with (85)'s bound, since concomitant phase is typically small), plus a finite-difference discretization error that depends on k-space grid resolution and was not present in the exact closed-form route above. What it buys is replacing $L_z \cdot Q_c$ *new NUFFT/kernel constructions* with $L_z \cdot Q_c$ *cheap stencil evaluations* on arrays already being computed — L_z NUFFTs (unchanged from the base operator) plus $L_z \cdot Q_c$ stencils, instead of $L_z \cdot Q_c$ NUFFTs. A genuine but more modest saving than eliminating the coupling itself, which (87) does not do.

4 Closed-Form Normal-Operator Compression, With Coils

The previous draft of this section wrongly framed the normal-operator compression as a 3-way tensor (CP) decomposition of $T[j, i, k] = g_{jk}\bar{g}_{ik}$, and concluded — correctly, for that specific (wrong) framing — that no closed form exists in general (Kolda & Bader 2009; de Silva & Lim 2008 ill-posedness). That critique does not apply to the actual problem here. The point made against it: once coil maps, whitening, and the k-sum are folded into one operator acting on the image, the right object to compress is not a free-floating 3-tensor — it is the already-fixed $L \times L$ temporal structure coming out of the forward SVD, and reducing *that* is an

ordinary matrix problem, with the coil dimension factoring out for free. This section redoes the derivation properly.

4.1 The full operator, exactly, before any compression

With the L-term forward SVD $g_{jk} \approx \sum_l \Upsilon_{jl}\Omega_{kl}$ (Eckart–Young optimal for g) and whitened coil maps $\tilde{s}_{cj}$, the exact (no further approximation) normal operator is

$$N_{ij} = \Xi_{ij} \sum_{l_1,l_2=1}^{L} \Upsilon_{jl_1}\bar{\Upsilon}_{il_2}\, h_{l_1l_2}(\mathbf{r}_j - \mathbf{r}_i), \qquad \Xi_{ij} = \sum_{c=1}^{C} \tilde{s}^*_{ci}\tilde{s}_{cj}, \qquad h_{l_1l_2} = \mathbf{F}_{\text{adj}}(\Omega_{\cdot l_1} \odot \bar{\Omega}_{\cdot l_2}), \tag{88}$$

which, multiplying out Ξ and collecting terms, is

$$N_{ij} = \sum_{c=1}^{C} \sum_{l_1,l_2=1}^{L} \left[\tilde{s}_{cj}\Upsilon_{jl_1}\right]\left[\tilde{s}^*_{ci}\bar{\Upsilon}_{il_2}\right] h_{l_1l_2}(\mathbf{r}_j - \mathbf{r}_i). \tag{89}$$

This is *exactly* the user's ansatz $\sum_p \sum_c B_{pc}\mathbf{F}^H C_p D_p \mathbf{F} A_{pc}$, with $p \leftrightarrow (l_1, l_2)$ (so $P_{\text{raw}} = L^2$, not yet reduced), $A_{pc}(j) = \tilde{s}_{cj}\Upsilon_{jl_1}$, $B_{pc}(i) = \tilde{s}_{ci}\Upsilon_{il_2}$, and $C_pD_p \leftrightarrow h_{l_1l_2}$ — zero approximation error beyond the original L-term forward SVD. This matches the already-derived "Full Normal Operator with Coil Maps" section above, just not yet compressed from L^2 down to a smaller P.

4.2 Coils do *not* factor out of the compression problem

An earlier draft of this section claimed the opposite — that the $L^2 \to P$ compression could be solved on $h_{l_1l_2}/\mathbf{M}(t_k)$ alone, coil-free, and then paired with every coil for free afterward. **That is wrong.** The target of compression is $N_{ij} = \Xi_{ij}M_{ij}$ (Hadamard product, $M_{ij} := \sum_{l_1,l_2} \Upsilon_{jl_1}\bar{\Upsilon}_{il_2}h_{l_1l_2}(\mathbf{r}_j - \mathbf{r}_i)$), so $N_{ij} - \hat{N}_{ij} = \Xi_{ij}(M_{ij} - \hat{M}_{ij})$. Minimising the *unweighted* $\sum_{ij}|M_{ij} - \hat{M}_{ij}|^2$ is a different optimisation problem from minimising the *actually relevant* $\sum_{ij}|\Xi_{ij}|^2|M_{ij} - \hat{M}_{ij}|^2$, and the two have different minimisers whenever Ξ_{ij} is non-constant — exactly like weighted vs. unweighted least squares disagree whenever the weights aren't flat. Ξ_{ij} (coil-sensitivity products) is essentially never constant. Solving the coil-free problem and pairing with coils afterward optimises the wrong objective.

Redone with the weight included. The loss is now

$$\mathcal{L} = \sum_{i,j,k} |\Xi_{ij}|^2 \left|G_{jik} - \textstyle\sum_p \Lambda_{kp}A_{jp}B_{ip}\right|^2, \qquad G_{jik} = g_{jk}\bar{g}_{ik}. \tag{90}$$

Λ still enters linearly, so it is still eliminable in closed form (Variable Projection still applies) — but $\mathcal{G}$ no longer factors into one Hadamard product of two plain Gram matrices. Expanding $\Xi_{ij} = \sum_c \tilde{s}^*_{ci}\tilde{s}_{cj}$ and defining the *coil-weighted* factors $A^{(c)}_{jp} := \tilde{s}_{cj}A_{jp}$, $B^{(c)}_{ip} := \tilde{s}^*_{ci}B_{ip}$, the stationarity condition for Λ becomes

$$\mathcal{G}_{pp'} = \sum_{c,c'} \left(A^{(c')H}A^{(c)}\right)_{pp'} \left(B^{(c')H}B^{(c)}\right)_{pp'}, \qquad \boldsymbol{\ell}(k)_p = \sum_{c,c'} F^{(c,c')}_{kp}\, C^{(c,c')}_{kp}, \qquad \Lambda(k) = \mathcal{G}^{-1}\boldsymbol{\ell}(k), \tag{91}$$

where $F^{(c,c')}_{kp} = \sum_j \tilde{s}_{cj}g_{jk}\bar{\tilde{s}}_{c'j}\bar{A}_{jp}$ and $C^{(c,c')}_{kp} = \sum_i \tilde{s}^*_{ci}\bar{g}_{ik}\tilde{s}_{c'i}\bar{B}_{ip}$ are coil-PAIR-weighted matvecs against $g/\bar{g}$. **This is still closed form — Variable Projection survives — but it is genuinely C^2-coupled**, not C-independent: $\mathcal{G}$ sums C^2 Hadamard products (not 1), and each ALS/VARPRO step needs C^2 coil-pair-weighted matvecs (not C). For a real receive array ($C \sim$ 8–32) this is a 64–1024× cost increase over the (wrong) coil-free version, and the optimal A, B, Λ now depend on the actual coil sensitivity maps — they are not portable across different coil arrays/subjects without recomputing, unlike the coil-free $\{h_{l_1l_2}\}$ compression. If Ξ's rank is reduced first (noise-covariance-whitened virtual coils, $Q < C$), $C^2 \to Q^2$, but the coupling does not disappear unless $Q = 1$ (a single effective channel, Ξ_{ij} then genuinely rank-1 and the naive Hadamard factorisation becomes exactly correct again).

4.3 The $L^2 \to P$ reduction is two exact matrix steps, not a hard tensor problem

Vectorize the upper-triangular entries of $\mathbf{M}(t_k)$ into a length-$\frac{L(L+1)}{2}$ vector for every k, stack these as columns of a $\frac{L(L+1)}{2} \times K$ matrix $\mathbf{Z}$, and take its rank-P SVD:

$$\mathbf{Z} \approx \sum_{p=1}^{P} s_p \, \mathbf{a}_p \mathbf{b}_p^H, \qquad \Lambda_{kp} := s_p \, a_p(t_k) \tag{92}$$

— an ordinary matrix SVD, Eckart–Young optimal for this $\frac{L(L+1)}{2} \times K$ matrix, full stop (this is exactly (27) above; the previous section's mistake was treating this as needing a joint 3-mode tensor factorization, when combining (l_1, l_2) into one flat row-index *first* — principled, since the output must be a Hermitian quadratic form, not an arbitrary choice — turns it into a plain matrix SVD). Reshape each $\mathbf{b}_p$ back into an $L \times L$ Hermitian matrix $C_{\cdot\cdot p}$ and eigendecompose it,

$$C_{l_1 l_2, p} \approx \sum_{r=1}^{R} \lambda_{rp} \hat{C}_{l_1 rp} \bar{\hat{C}}_{l_2 rp}, \qquad \Theta_{j,rp} := \sum_{l} \hat{C}_{lrp} \, \Upsilon_{jl} \tag{93}$$

— again an ordinary, exact eigendecomposition of a tiny $L \times L$ Hermitian matrix, one per mode p. Both steps are closed form. **There is no hard tensor problem anywhere in this pipeline** — it never requires simultaneously fitting independent factors along three modes (which would be the genuinely hard CP/Tucker problem); it sequentially fixes one matrix problem at a time, each exactly solvable, because the (l_1, l_2) pairing is collapsed into a single axis before the first SVD rather than left as a free tensor mode.

The one honest caveat. Solving (92) then (93) *sequentially* is optimal at each individual stage, but is not guaranteed to be *jointly* optimal over the combined (P, R) pair simultaneously — the same mild, well-understood gap as sequential HOSVD versus the true optimal Tucker decomposition (de Lathauwer, de Moor, Vandewalle, *A Multilinear Singular Value Decomposition*, SIAM J. Matrix Anal. 2000), which comes with a *proven, bounded* suboptimality factor — not the unbounded/ill-posed CP situation wrongly invoked in the previous draft.

4.4 Eliminating Λ exactly: variable projection

The reduction above still left a residual gap (sequential, not jointly optimal). Push further by writing the optimality conditions of the L^2 least-squares problem directly and solving them, rather than appealing to a heuristic initialisation for ALS. Drop the orthonormality constraints on A, B (they were a simplifying assumption, not a requirement — the redundant C_p, D_p pair is absorbed into a single Λ_p as already noted), and minimise

$$\mathcal{L}(A, B, \Lambda) = \sum_{j,i,k} \Big| G_{jik} - \sum_{p} \Lambda_{kp} A_{jp} B_{ip} \Big|^2, \qquad G_{jik} := g_{jk} \bar{g}_{ik}. \tag{94}$$

Λ enters (94) *linearly* for fixed A, B — this is a separable nonlinear least squares problem, and Λ can be eliminated in closed form exactly (the Variable Projection method of Golub & Pereyra, *The Differentiation of Pseudo-Inverses and Nonlinear Least Squares Problems Whose Variables Separate*, SIAM J. Numer. Anal. 1973), leaving a smaller nonlinear problem in A, B alone — a real reduction, not a heuristic.

The restriction to $\mathrm{span}\{\Upsilon\}$ **is exact, not a simplifying assumption (coil-free case only).** Before substituting, justify why this restriction loses nothing. $g_{jk} = \sum_l \Upsilon_{jl} W_{kl}$ means g's j-fibre lies in $\mathrm{span}\{\Upsilon\}$ by construction — not an assumption, a direct consequence of g already being an L-rank factorisation. Hence $G_{jik} = g_{jk}\bar{g}_{ik}$ has *zero* component outside $\mathrm{span}\{\Upsilon\} \otimes \mathrm{span}\{\Upsilon^*\}$ for every k — there is nothing in G living outside that subspace to approximate. Decompose any candidate $A = A^{\parallel} + A^{\perp}$ (projection onto $\mathrm{span}\{\Upsilon\}$ plus orthogonal remainder), likewise B. Since $\Upsilon^{\perp} \perp \Upsilon$ exactly, the four cross terms of $\sum_p \Lambda_{kp} A_{jp} B_{ip}$ $(\|\|, \|\perp, \perp\|, \perp\perp)$ are mutually orthogonal in the *unweighted* Frobenius inner product, and G only overlaps the $\|\|$ piece. Pythagoras gives

$$\|G - \textstyle\sum_p \Lambda AB\|^2 = \|G_{\|\|} - (\sum \Lambda AB)_{\|\|}\|^2 + \|(\sum \Lambda AB)_{\|\perp}\|^2 + \|(\sum \Lambda AB)_{\perp\|}\|^2 + \|(\sum \Lambda AB)_{\perp\perp}\|^2. \tag{95}$$

The last three terms are pure added error — zero benefit, since G has nothing there to match. So the *global* minimiser over **all** $[N, P]$ matrices necessarily has $A^{\perp} = 0, B^{\perp} = 0$. Restricting to span$\{\Upsilon\}$ is exactly where the unconstrained optimum already sits — provably lossless, not a tradeoff, *for the coil-free (C=1, or uniform-coil) loss* (94) *as written.*

This does not extend to coil weighting. The coil-weighted loss (§4.2) is $\sum_{j,i,k} |\Xi_{ij}|^2 |M_{jik} - \sum_p \Lambda_{kp} A_{jp} B_{ip}|^2$ — weighted by $|\Xi_{ij}|^2$, which depends on (i, j) in a way that does *not* respect the $\Upsilon/\Upsilon^{\perp}$ split in general. The cross terms in (**??**) relied on $\Upsilon^{\perp} \perp \Upsilon$ in the *unweighted* inner product; under $|\Xi_{ij}|^2$-weighting that orthogonality does not carry over, so the cross terms no longer vanish for an arbitrary coil pattern. Restricting $A^{\text{core}}, B^{\text{core}}$ to span$\{\Upsilon\}$ in `als_refine_coil_weighted` is therefore a practical, tractable choice once coils are non-uniform — **not** a proven exact equivalence. This is the same fact already established in §4.2 ("coils do not factor out of the compression problem") recurring here in a different guise — consistent with it, not a new result.

Returning to the coil-free case: A, B are not arbitrary $[N, P]$ matrices in the implementation, they are constrained to $A_{jp} = \sum_l \Upsilon_{jl} A^{\text{core}}_{lp}$, $B_{ip} = \sum_{l'} \Upsilon^*_{il'} B^{\text{core}}_{l'p}$ — linear combinations of the existing L orthonormal spatial modes, with the core coefficients $A^{\text{core}}, B^{\text{core}}$ ($[L, P]$) being what is actually optimised. Substituting these and $g_{jk} = \sum_l \Upsilon_{jl} W_{kl}$ into (94),

$$G_{jik} - \sum_p \Lambda_{kp} A_{jp} B_{ip} = \sum_{l,l'} \Upsilon_{jl} \Upsilon^*_{il'} M_{ll'k}, \qquad M_{ll'k} := W_{kl} \bar{W}_{kl'} - \sum_p A^{\text{core}}_{lp} \Lambda_{kp} B^{\text{core}}_{l'p}. \tag{96}$$

Summing $|\cdot|^2$ over j, i and using orthonormality $\sum_j \Upsilon_{jl} \Upsilon^*_{jm} = \delta_{lm}$ twice,

$$\sum_{j,i} \Big| \sum_{l,l'} \Upsilon_{jl} \Upsilon^*_{il'} M_{ll'k} \Big|^2 = \sum_{l,l',m,m'} M_{ll'k} \bar{M}_{mm'k}\, \delta_{lm}\, \delta_{l'm'} = \sum_{l,l'} |M_{ll'k}|^2, \tag{97}$$

exactly — not an approximation, not a Fourier/Parseval argument, just $\Upsilon^H \Upsilon = I_L$ (true because Υ is SVD-derived). Therefore

$$\mathcal{L}(A, B, \Lambda) = \sum_{l,l',k} \Big| W_{kl} \bar{W}_{kl'} - \sum_p A^{\text{core}}_{lp} \Lambda_{kp} B^{\text{core}}_{l'p} \Big|^2, \tag{98}$$

a loss over (l, l', k) only — $[L, L, K]$-sized, no voxel index j, i anywhere. **This is why the ALS/VARPRO minimisation, and the diagnostic loss `varpro_reduced_loss` that tracks it, never construct a single voxel- or grid-resolution Toeplitz kernel**: the N-voxel sums in the original (94) have already been summed out in closed form via (96) before the optimisation ever runs, leaving an equivalent problem entirely in core (L, P, K-sized) space. Sanity check against (97) with $A^{\text{core}} = B^{\text{core}} = 0$: $\sum_{l,l',k} |W_{kl} \bar{W}_{kl'}|^2 = \sum_k \big(\sum_l |W_{kl}|^2\big)^2$ — exactly `g_norm_sq`, validated bit-for-bit against a brute-force voxel-space computation in `test_gnorm_closed_form`. The Toeplitz spatial kernels ($h_{ll'}$, built per (l, l') pair via the sampling pattern) are a separate, downstream representation used only to *apply* whichever operator (naive L^2-term or compressed P-term) is ultimately deployed — the fitting/compression step that decides $A^{\text{core}}, B^{\text{core}}, \Lambda$ never needs them.

Stationarity in Λ. Setting $\partial \mathcal{L} / \partial \Lambda^*_{kp} = 0$ for each k gives the normal equations

$$\sum_{j,i} G_{jik}\, \bar{A}_{jp} \bar{B}_{ip} = \sum_{p'} \Lambda_{kp'} \underbrace{\Big(\sum_j A_{jp'} \bar{A}_{jp} \Big)}_{\mathcal{A}_{pp'}} \underbrace{\Big(\sum_i B_{ip'} \bar{B}_{ip} \Big)}_{\mathcal{B}_{pp'}}, \tag{99}$$

since the (j, i)-indexed "design vector" for mode p' is the outer product $A_{\cdot p'} \otimes B_{\cdot p'}$, whose Gram matrix factors into the *Hadamard product* of the separate Gram matrices $\mathcal{A} = A^H A$, $\mathcal{B} = B^H B$ (a standard fact for Khatri-Rao/Kronecker-structured design matrices). Writing $\boldsymbol{\ell}(k)_p := \sum_{j,i} G_{jik} \bar{A}_{jp} \bar{B}_{ip}$ and $\mathcal{G} := \mathcal{A} \odot \mathcal{B}$ (elementwise product, $P \times P$), (98) is exactly $\boldsymbol{\ell}(k) = \mathcal{G}\, \boldsymbol{\Lambda}(k)$, so

$$\boxed{\boldsymbol{\Lambda}(k) = \mathcal{G}^{-1} \boldsymbol{\ell}(k)} \tag{100}$$

whenever $\mathcal{G}$ is invertible — a genuine closed form for Λ given A, B, with no orthonormality assumption. **This is a strict generalisation of the document's existing $\Lambda = F \odot C$ update**: when A, B are constrained orthonormal ($\mathcal{A} = \mathcal{B} = I_P$), $\mathcal{G} = I_P \odot I_P = I_P$ exactly, and (99) collapses to $\Lambda_{kp} = \ell(k)_p$, recovering $\Lambda = F \odot C$ — so the orthonormal case was never doing anything different, it was just the special case where $\mathcal{G}$ happens to be the identity.

The reduced problem. Substituting (99) back into (94) (the standard "profiling out a linear nuisance parameter" identity) gives a loss depending on A, B alone,

$$\mathcal{L}(A, B) = \|G\|_F^2 - \sum_k \boldsymbol{\ell}(k)^H \, \mathcal{G}(A, B)^{-1} \, \boldsymbol{\ell}(k), \tag{101}$$

i.e. minimising (94) over all three factors jointly is *exactly equivalent* to maximising the generalised-Rayleigh-quotient-like quantity $\sum_k \boldsymbol{\ell}(k)^H \mathcal{G}(A, B)^{-1} \boldsymbol{\ell}(k)$ over A, B only. This is a real reduction from a 3-block to a 2-block problem — one fewer coupled factor, with the elimination exact rather than approximate.

Why this is where it stops. $\boldsymbol{\ell}(k)$ is bilinear in (A, B) and $\mathcal{G}$ is biquadratic in (A, B), so (100) is still genuinely nonlinear and non-convex in the remaining (A, B) — eliminating Λ removes exactly the one block that entered *linearly*; A and B enter the original model multiplicatively with each other (through $A_{jp}B_{ip}$) and cannot both be eliminated the same way (fixing Λ and A makes B linear and eliminable too, by the identical argument — but then A no longer is, and vice versa: only ever one block can be profiled out in closed form at a time). This is not a new gap — it is the same fact already cited (no Eckart–Young analogue once three or more factors are coupled multiplicatively), now *localised* to exactly the (A, B) pair rather than the whole problem. **Net result: a provably exact, non-heuristic two-block reduction ((99)–(100)), with the genuinely hard part of the problem isolated to the smallest residual core it can be — Variable-Projection-accelerated ALS on (A, B) only, rather than naive 3-block ALS on (A, B, Λ).** This is a real, citable improvement to the existing algorithm, not a restatement of the impossibility result.

4.5 Full algorithm

Stage 0 — coil-free closed-form initialisation (replaces the ad hoc top-P-pair one-hot init):

1. From the L-term forward SVD $(\Upsilon, \Omega, \boldsymbol{\sigma})$, form $\mathbf{M}(t_k) = \boldsymbol{\Omega}_{k,:}\boldsymbol{\Omega}_{k,:}^H \in \mathbb{C}^{L \times L}$ for every sample k (rank-1, Hermitian, by construction).
2. Vectorize the upper-triangular entries of $\mathbf{M}(t_k)$ into a $\frac{L(L+1)}{2}$-vector, stack over k into $\mathbf{Z} \in \mathbb{C}^{L(L+1)/2 \times K}$, take its rank-$P$ SVD (92).
3. Reshape each right singular vector back into an $L \times L$ Hermitian matrix and take its *full* eigendecomposition (all L eigenpairs, not just the dominant one – a generic $L \times L$ Hermitian matrix spreads energy across several eigenvalues, and $R = 1$ measurably underperforms keeping all of them, see below). Pool candidates across *every* retained $\mathbf{Z}$-mode $\times$ every eigenvector and keep the global top-$P \cdot R$ by score $|s_p \cdot \lambda_{rp}|$ — this lets a single dominant $\mathbf{Z}$-mode contribute several terms rather than forcing exactly one eigenvector per mode.
4. Set $A^{(0)}_{j,(r,p)} = \sqrt{|\lambda_{rp}|} \sum_l \hat{C}_{lrp} \Upsilon_{jl}$, $B^{(0)}_{j,(r,p)} = \mathrm{sign}(\lambda_{rp}) \sqrt{|\lambda_{rp}|} \sum_l \hat{C}_{lrp} \Upsilon_{jl}$ — the sign correction on $B^{(0)}$ is necessary, not optional: these Hermitian matrices are generally *indefinite* (not PSD), so $\sqrt{|\lambda|}$ on both sides silently drops the sign whenever $\lambda_{rp} < 0$.

Empirical validation (`normal_toeplitz_als_test.cpp`). Pre-ALS, this init's loss is occasionally slightly *worse* than the old top-P-pairs heuristic (the new candidates, drawn from different $\mathbf{Z}$-modes, are not mutually orthogonal, so picking the global top-$P \cdot R$ by individual score can pay a redundancy cost the old one-hot init never incurs). What matters is the *converged* result: running the same ALS refinement (Stage 1) from both inits, this init reaches a measurably lower final loss after $\sim$20 iterations, on every random draw tested — confirming it as the better choice in the regime that matters (the production call site already runs 50 ALS iterations).

Stage 1 — VARPRO-style refinement (closed-form Λ every step, no orthonormality assumed — this is what the document's existing ALS code already does structurally for the Λ step; the gap is only in initialisation and convergence tracking):

Repeat until $\mathcal{L}(A, B)$ (100) stops decreasing (relative tolerance, e.g. 10^{-4}) or a max-iteration budget is hit:

1. Form $\mathbf{F} = \mathbf{W}\bar{A}$, $\mathbf{C} = \bar{\mathbf{W}}B$ where $\mathbf{W} = \Omega \odot S \in \mathbb{C}^{K\times L}$ (matrix-free against the dense $[L, P \cdot R]$ core — no N- or K-scale tensor ever formed, matching the existing code's core-space trick).
2. $\boldsymbol{\ell}(k) = \mathbf{F}_{k,:} \odot \mathbf{C}_{k,:}$ for every k; $\mathcal{G} = (A^H A) \odot (B^H B)$; solve $\Lambda(k) = \mathcal{G}^{-1}\boldsymbol{\ell}(k)$ (99) (truncated pseudo-inverse if $\mathcal{G}$ is ill-conditioned, as the existing `nh_rsolve_herm` already does).
3. Evaluate $\mathcal{L}(A, B)$ via (100) — **record this**; it is currently not tracked at all (fixed iteration count, no stopping criterion).
4. Update A given (B, Λ) closed form (linear in A for fixed B, Λ — identical structure to the existing A-update); update B given (A, Λ) symmetrically.
5. *(Stretch goal, not yet derived in closed form here)* True VARPRO additionally corrects the A- and B-updates for Λ's own dependence on A, B via the implicit-function chain rule through (99) (the standard Golub–Pereyra projected-Jacobian correction) — this is what separates VARPRO from plain alternating block coordinate descent in the literature, and is the one piece of the method not yet worked out term-by-term here.

Stage 2 — coils. Not free (Section 4.2): naively pairing every mode with every coil ($A_{p'c} = A_{p'} \odot \tilde{s}_c$, etc.) reuses Stages 0–1's Λ, which solved the wrong (coil-free) objective. The correct version reruns Stage 1's VARPRO loop with the C^2-coupled $\mathcal{G}/\boldsymbol{\ell}(k)$ of (91) — Stage 0's closed-form init can still seed A, B (a reasonable starting point, just not the converged answer), but Λ must be re-solved coil-aware before trusting the result. $P \cdot R$ Toeplitz kernels, $P \cdot R \cdot C$ diagonal pairs in the final applied operator either way — the cost structure target is unchanged, only how Λ is *found* changes.

5 Gradient Nonlinearity via Coordinate Substitution (Warp)

5.1 Separability assumption

Split the higher-order field terms in $\Delta B(\mathbf{r}, t)$ into two classes:

- **Static GNL terms**: spatial harmonics $N_l^m(\mathbf{r})$ ($l \geq 2$) whose time waveform is, to good approximation, the *same* impulse response as the linear term of the physical gradient coil that produced them, $G_\beta^{lm}(t) \approx c_\beta^{lm}\, G_\beta^0(t)$. These are pure coil-geometry imperfections — fixed spatial pattern, scaled by the same coil current as the linear gradient.
- **Residual terms**: $\Delta B_0(\mathbf{r})$ (off-resonance) and concomitant fields, whose time dependence is quadratic in the gradient waveforms (e.g. $G_x(t)G_z(t)$) and therefore cannot share a single coil's linear waveform.

Define, per axis β, the composite coordinate field

$$u_\beta(\mathbf{r}) = r_\beta + \sum_{\substack{l\geq 2 \\ \text{static}}} c_\beta^{lm} N_l^m(\mathbf{r}), \qquad \mathbf{u}(\mathbf{r}) = (u_x, u_y, u_z). \tag{102}$$

Under the separability assumption,

$$\sum_\beta G_\beta^0(t) r_\beta + \sum_{\text{static GNL}} G_\beta^{lm}(t) N_l^m(\mathbf{r}) = \sum_\beta G_\beta^0(t)\, u_\beta(\mathbf{r}) = \mathbf{k}(t) \cdot \mathbf{u}(\mathbf{r}). \tag{103}$$

What remains in $\Phi_{\text{res}}(\mathbf{r}, t)$ is only T_2 decay, ΔB_0, and concomitant fields — i.e. Φ from (10) with the static-GNL channels removed from Q.

5.2 Coordinate substitution

With $\mathbf{q} = \mathbf{u}(\mathbf{r})$,

$$\begin{aligned} S_c(t) &= \int_V s_c(\mathbf{r})\rho(\mathbf{r}) e^{\Phi_{\text{res}}(\mathbf{r},t)} e^{-i\mathbf{k}(t)\cdot\mathbf{u}(\mathbf{r})}\, d\mathbf{r} \\ &= \int_{\mathbf{u}(V)} \underbrace{s_c(\mathbf{r}(\mathbf{q}))\rho(\mathbf{r}(\mathbf{q}))\,|\det \mathbf{J}_{\mathbf{r}}(\mathbf{q})|}_{\tilde{\rho}_c(\mathbf{q})}\; e^{\Phi_{\text{res}}(\mathbf{r}(\mathbf{q}),t)} e^{-i\mathbf{k}(t)\cdot\mathbf{q}}\, d\mathbf{q}, \end{aligned} \tag{104}$$

where $\mathbf{J}_{\mathbf{r}} = \mathbf{J}_{\mathbf{u}}^{-1}$. This is exactly the $\mathbf{A}^H\mathbf{A}$ Toeplitz formalism of the previous section, evaluated in $\mathbf{q}$-coordinates, with Φ_{res} now spanning only off-resonance and concomitant fields (fewer histogram channels Q, no static-GNL blow-up of n_{hist}).

5.3 Two-grid discretisation

- **q-grid**: regular Cartesian, chosen at FFT-optimal size — the grid the Toeplitz kernel (57) is padded/transformed on, unchanged from the existing implementation.
- **r-grid**: physical voxel grid (image domain, where the regularizer lives), *oversampled* relative to the nominal resolution so that $\mathbf{u}$(**r**-grid domain) $\supseteq$ **q**-grid domain — i.e. no extrapolation in $\mathbf{W}$ below. The **r**-grid is never FFT'd, so its size is unconstrained by factorization concerns.
- $\mathbf{W}$: sparse linear interpolation **r**-grid $\to$ **q**-grid representing $\mathbf{q} = \mathbf{u}(\mathbf{r})$, precomputed once (static, determined by gradient-coil hardware). $\mathbf{W}^H$ is its adjoint, **q**-grid $\to$ **r**-grid.

5.4 Jacobian weighting folded into coil maps

$\sqrt{|\det \mathbf{J}_{\mathbf{u}}(\mathbf{r})|}$ is a diagonal, closed-form (polynomial-in-**r**, since $\mathbf{u}$ is polynomial) scaling, precomputed once and absorbed into effective coil sensitivities,

$$s_c'(\mathbf{r}) = s_c(\mathbf{r})\,\sqrt{|\det \mathbf{J}_{\mathbf{u}}(\mathbf{r})|}. \tag{105}$$

This keeps $\mathbf{W}$ close to a partial isometry ($\mathbf{W}^H\mathbf{W} \approx \mathbf{I}$ on the overlap region), preserving adjoint consistency and CG conditioning without extra operator applications.

5.5 Diffeomorphism requirement

$\mathbf{u}$ must be invertible ($\det \mathbf{J}_{\mathbf{u}} \neq 0$) over the oversampled **r**-grid domain. $\mathbf{J}_{\mathbf{u}} = \mathbf{I}$+(small polynomial perturbation), whose magnitude grows with $|\mathbf{r}|$ (the quadratic GNL terms scale as $\alpha\,|\mathbf{r}|$ in the Jacobian). This is *not* assumed a priori — verify $\det \mathbf{J}_{\mathbf{u}}(\mathbf{r}) > 0$ numerically on the **r**-grid (cheap, closed form). If it fails near the FOV edge for a given coefficient set, either crop the **r**-grid domain or leave the offending high-order term in Φ_{res} instead of absorbing it into $\mathbf{u}$.

5.6 Resulting operator chain

$$\mathbf{A}_c = \mathbf{F}_{\mathbf{q}}\,\mathbf{D}_{\text{res}}\,\mathbf{W}\,\text{diag}(s_c'), \tag{106}$$

$$\mathbf{A}_c^H\mathbf{A}_c = \text{diag}(s_c'^{*})\,\mathbf{W}^H\,\mathbf{D}_{\text{res}}^H\,(\text{Toeplitz}_{\mathbf{q}})\,\mathbf{D}_{\text{res}}\,\mathbf{W}\,\text{diag}(s_c'), \tag{107}$$

where $\text{Toeplitz}_{\mathbf{q}}$ is (60) unchanged — $\mathbf{k}(t)\cdot\mathbf{q}$ is linear and translation-invariant on the regular **q**-grid, so the PSF/Toeplitz embedding is unaffected by the warp. $\mathbf{D}_{\text{res}}$ is the existing low-rank histogram/SVD interpolator, now over off-resonance + concomitant fields only. $\mathbf{W}$, $\mathbf{W}^H$ add one sparse interpolation each per normal-operator application — $O(N)$, negligible next to the Toeplitz FFTs. Image-domain regularizers are applied in **r**-space each iteration (i.e. $\mathbf{W}$ sits *inside* the iterative loop, not just as a final resampling step).